

new equipment and makes utilisation of a variety of cloud applications possible that are otherwise cost-prohibitive to purchase.

Cloud computing provides the capability to shift from on-premises data solutions to colocation or cloud-based services. The distributed nature of cloud technology makes it accessible to multiple organisations as the data and applications stored in a cloud network can be accessed by anyone who has the right to access it. This permits transparency and data sharing within a secure framework.

The ability to share and move data easily provides interoperability and rapid knowledge sharing. Cloud computing enables healthcare professionals working in the field to quickly and easily transmit on-site data to a shared location, which can be accessed instantly by the researchers. Leading cloud computing organisations have been proactively supporting cloud-based healthcare research. The US Centres for Disease Control and Prevention (CDC) used cloud technology to combat influenza in the wake of the 2009 H1N1 outbreak and also found it highly effective for viral research during West Africa's Ebola crisis. IBM has now made available a number of cloud computing resources for free to Covid-19 researchers to model the spread of the disease and identify drugs for effective treatments. Amazon Web Services (AWS) has provided twenty million dollars in cloud credits to subsidise research into diagnostic tools related to Covid-19.

Oracle is actively participating in coronavirus vaccine development by providing computing platforms for clinical trials. Oracle has also introduced Oracle Clinical Trials Systems to gather data on Covid-19 drug testing and therapeutic learning to serve as a repository for all Covid-19 treatments being administered.



Oak Ridge National Laboratory's Summit—The world's fastest supercomputer is being used for Covid-19 research (Credit: ORNL photo from www.geekwire.com)

Artificial intelligence and heuristic methods

Artificial intelligence (AI) and heuristic methods, in particular, machine learning, data mining, cluster analysis, pattern recognition, and knowledge representation, are helping meet the challenges posed by the Covid-19 pandemic. AI helps in reducing the workload of healthcare workers by tracking, screening, early detection, and diagnosis of the infection. It helps researchers track the spread of the virus and predicts mortality risk by analysing the data of the patients.

AI helps in identifying high-risk patients by population screening, providing medical help, making notifications, and controlling the infection. The flow diagram on next page compares the AI and non-AI based procedures followed to identify the Covid-19. The AI-based procedure provides higher accuracy and reduces complexity, time taken, and the number of steps needed.

Key areas where the AI approach has been found encouraging and fruitful are knowledge acquisition from genetic data, interpretation of nucleotide sequences, prediction of protein's structure and function, and drug design. AI promptly analyses the irregular symptoms and other 'red flags' present in patients to help

healthcare functionaries make faster and cost-effective decisions. It helps in identifying the clusters and 'hot spots' of infection, predicting the future course of the disease and chances for its reappearance, contact tracing of the infected individuals, and providing appropriate treatments.

Application of GIS to Covid-19

Geographic understanding is essential in detecting, understanding, and responding to pandemics such as Covid-19. A geographic information system (GIS) is a computer system for capturing, storing, checking, and displaying data related to positions on the earth's surface in a meaningful way through interactive maps or other infographics. GIS techniques and tools are useful in finding and monitoring large-scale population-based occurrences such as disease clusters, outbreaks of infection, and possible associations between infections and environmental factors.

GIS helps public health agencies, policymakers, and administrators to map disease occurrence against multiple parameters such as demographics, environment, geographies, and past occurrences to understand the origin of an outbreak, its spread pattern, and intensity. GIS enables identifying at-risk populations, planning targeted intervention, evaluating facilities and healthcare capacities, and establishing effective communication among supporting agencies to ensure a coordinated response.

It helps epidemiologists study the outbreak of diseases and how the incidence of congenital diseases varies over space and time. WHO, Red Cross, academics, local governments, and hospitals employ GIS to communicate spatial data sets to the public and help